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# Alfalfa in arid and semi-arid regions Taroudant as an example, good and bad cultural practices on the environment: A statistical study

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**Abstract:** A field study was made on the cultivation of alfalfa and their different components of this crop in the Taroudant region in the south of Morocco, the questions of this study based on the cultivation and fertilization methods, irrigation, and finally prospects. This study showed that the majority will continue to cultivate alfalfa her because the use of alfalfa is mainly as fodder food for their own herd, the whole population does the plowing before the crops, the varieties used are mainly from USA. The population uses manure with different quantities, it depends on the cultivated area and on the use of chemical fertilization. This study showed high consumption of water and chemical fertilizers with a means between 44066.8 m<sup>3</sup>/year/ha is 62399.9 m<sup>3</sup>/year/ha, and between 165.6 kg/ha/year and 205.7 Kg/ha/year respectively with a 95% confidence interval. What knocks the alarm danger in environmental matters.

**Keywords:** Alfalfa, Environment, Fertilization, Irrigation, Taroudant, Morocco.

## 1. Introduction

Agriculture in Morocco is the fastest growing sector in the country, this sector including livestock currently represents about 15% of the gross domestic product of Morocco, and almost a quarter of its exports [1], which requires providing suitable feed to guarantee the success of the overall ration in order to maintain stable animal production [2], for that a diet which is constituted a proportion of alfalfa increases the value of this diet taking into account its low production inputs, its environmental and nutritional benefits and its continuous production for four to five years [3].

The problem of drought and the increased water needs by agriculture in Morocco [4], requires to characterize this primordial crop in order to solve it, indeed the crop issue, the needs hydric, nutritional and pedoclimatic, are the main causes for a bad cultural practice which destroyed the soil. Furthermore, the environment and the natural resources used in the region decreased the quality of soil



as a result the decrease of plant growth. Our study focused to draw the next perspectives related to the continuity of the crop of alfalfa in the conditions current by a questionnaire.

## **2. Material and methods**

### *2.1. Sampling method and sample size*

The survey carried out in the south of Morocco in region of Taroudant which are recognized by agriculture. The survey was carried out on a total sample of one hundred farms. In fact; we worked on a five regions with a sample size of thirteen to fifteen percent of the total population growing alfalfa.

### *2.2. Questionnaire and survey compositions*

The questionnaire was in the form of a survey focused on the components, objectives, and prospects for growing alfalfa. It was divided into three parts. Firstly, the questions describe the population, the land and the objectives of cultivation. Secondly, the method of growing alfalfa by the description of the tillage practice, the amendments, the sowing, the irrigation as well as the harvest and the diseases of this crop. Finally, the perspectives in the form of questions on the continuity of cultivation of alfalfa.

### *2.3. Data analysis*

The statistical analyses were performed using XLSTAT.v2018. The direction of different relationships existing between the different nominal qualitative variables is done using the MCA method and for the quantitative variables we used the PCA method. The modes are also calculated for all the components, the quantiles and the medians are schematized in the form of a Boxplot. The estimate of the average differences is made by a 95% confidence interval by IBM SPSS Statistics 23.

## **3. Results and discussion**

### *3.1. Response rates*

The results of the survey by which we have based in this study were only eighty-five agricultures of one hundred surveys; indeed, four farmers refused the response of the survey. While eleven questionnaires were eliminated since it was no answer on certain questions of our survey.

### *3.2. Description of the population*

The results of the descriptive statistics Table 1 show that alfalfa is mainly used as a fodder food for herds this represents 96%, and also shows that the most dominant complementary food ratio is Corn silage + Straw + Cereals which represents more than 56% of all dietary ratios, farmers do not depend on pasture in diets since this represents a small proportion.

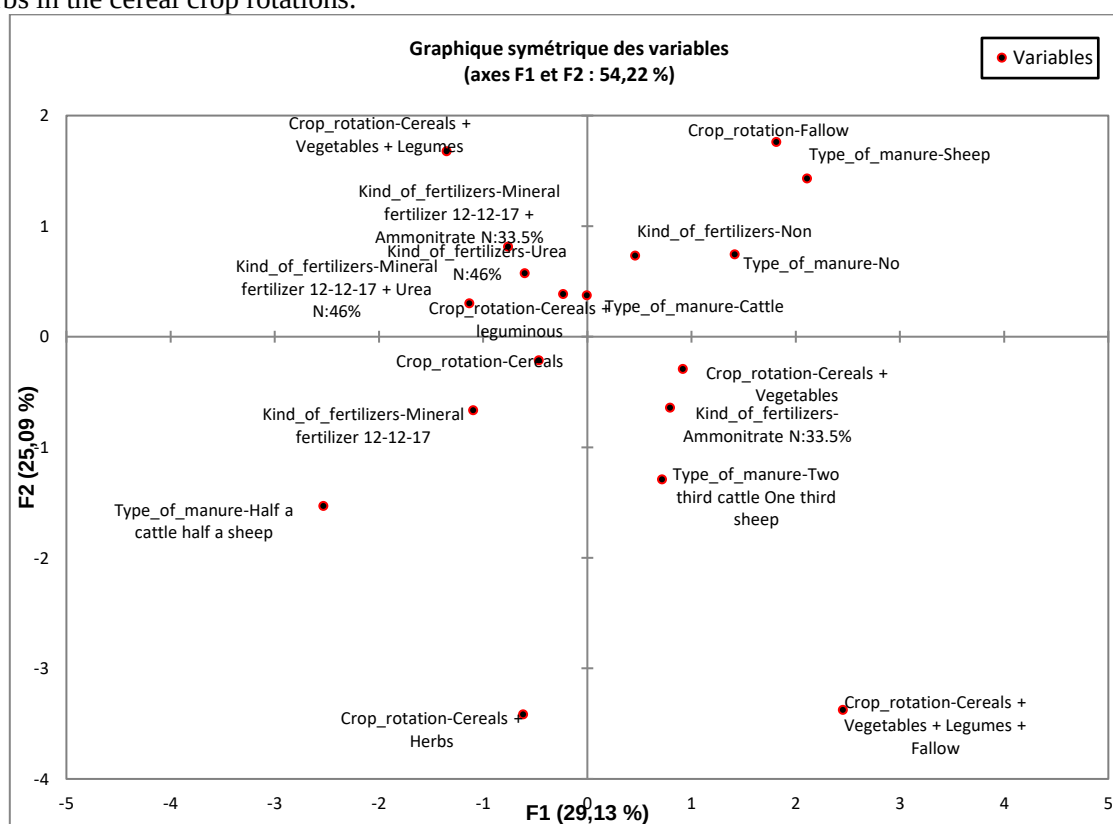
The diversification of supplementary fodder regime for alfalfa in the population due to the availability of these foods for agriculture and the food needs of the herd. The nutrient content of many alternative foods varies widely, making it necessary to analyze or assess the value of the food. Producers need to know the energy, protein and key levels of minerals in these foods to develop a balanced, low-cost diet for livestock to meet the needs of livestock and the objectives of producers, the ration must be balanced [5]. From a commercial point of view, alfalfa is among the largest forage crops in many countries. For all that, forage legumes can also be high quality animal feed. But its growth is threatened by changing climatic conditions, which indicates the need to cultivate varieties capable of resisting and also adapting to the negative effects of climate change [6].

**Table 1.** The descriptive statistics of the farming practice of alfalfa.

| Variable\Statistics       | Nb. of observations | Mode                      | Mode (effective) | Modalities                            | Proportion by modality |
|---------------------------|---------------------|---------------------------|------------------|---------------------------------------|------------------------|
| <b>Use_of_alfalfa</b>     | 85                  | fodder for personal herd  | 82               | Sale                                  | 0.035                  |
|                           |                     |                           |                  | fodder for personal herd              | 0.965                  |
|                           |                     |                           |                  | Corn silage+Green beans+Straw+Cereals | 0.153                  |
|                           |                     |                           |                  | Corn silage+Straw+ Cereals            | 0.565                  |
|                           |                     |                           |                  | Corn silage+Straw+Cereals             | 0.188                  |
| <b>Complementary_food</b> | 85                  | Corn silage+Straw+Cereals | 48               | +Pasture                              | 0.024                  |
|                           |                     |                           |                  | Cereals                               | 0.024                  |
|                           |                     |                           |                  | Pasture                               | 0.012                  |
|                           |                     |                           |                  | Pasture+Green beans+Cereals           | 0.012                  |
|                           |                     |                           |                  | Sale                                  | 0.035                  |

### 3.3. Alfalfa cultivation method

The variables analyzed by the MCA are Type of manure; Kind of fertilizers; Crop rotation. The symmetric graphic outputs of the variables (Figure 1) visualize two principals groups of the methods of cultivation of alfalfa the first is based on the use of multiple types of manure, mineral fertilizers and multiple crop rotations at the same time, the second is based on the use of 33% ammonitrate, the use of a mixture of manure from two third cattle and one third sheep and a crop rotation based on cereals and vegetables, The other two modes are very rare either do not use any fertilizer and no crop rotation or a fertilizer with widely different NPK proportions to the first two modes, with the incorporation of herbs in the cereal crop rotations.



**Figure 1.** The symmetric graphic outputs of the Multiple Correspondence Analysis (MCA) for the variation of method of cultivation of alfalfa by agricultures of region. The variables analyzed: Type of manure; Kind of fertilizers; Crop rotation. The black dots indicate the variables studied.

The results of the descriptive statistics Table 2 show that the majority of the population uses Monosoc Brabant plow and Cover crops V, and anomalies that Two-soc Brabant plow and Cover crops PXL use in plowing, the entire population studied applied plows deeply followed by plowing superficial. Cattle manure occupies a large part of the organic amendments of this crop it is generally applied by the spreader followed by the proportions of cattle and sheep manure a very small proportion which uses sheep manure alone or does not use the manure, the unavailability of sheep manure is the reason for this low proportion of use, almost a third of the population studied uses Ammonitrate N: 33.5% as chemical fertilizers each quarter, the use of the Mineral fertilizer 12-12-17 when applying manure remains high compared to the use of Urea N: 46%, the use of Mineral fertilizer 12-12-17 + Ammonitrate N: 33.5% or Mineral fertilizer 12-12 -17 + Urea N: 46% do not even represent 5% of the population, almost 30% of the population does not use fertilizers.

Cereals and also cereals are found in all combinations of rotation crops except a proportion of 3% which leaves the land fallow, more than half of the fields studied suffer from diseases (Butterfly of *Cydia medicaginis* - *Tychius aureolus* - *Acyrtosiphon pisum*; Aphid - *Uromyces striatus*; rust) and 84% of the agricultural crops treat with chemicals these pests the remaining part either it does not treat or it uses conventional method to reduce the impact of pests on the crop (irrigation by the Water Diffuser or reduce the time between harvests).

**Table 2.** Descriptive statistics for the description of the population

| Variable\ Statistics                               | Nb. of observations | Mode                 | Mode (effective) | Modality                                          | Proportion by modality |
|----------------------------------------------------|---------------------|----------------------|------------------|---------------------------------------------------|------------------------|
| <b>Deep_plowing</b>                                | 85                  | Monosoc Brabant plow | 82               | Monosoc Brabant plow                              | 0.965                  |
| <b>Superficial_plowing</b>                         | 85                  | Cover crops V        | 82               | Two-soc Brabant plow                              | 0.035                  |
|                                                    |                     |                      |                  | Cover crops PXL                                   | 0.035                  |
|                                                    |                     |                      |                  | Cover crops V                                     | 0.965                  |
|                                                    |                     |                      |                  | Cattle                                            | 0.694                  |
|                                                    |                     |                      |                  | Half a cattle half a sheep                        | 0.035                  |
| <b>Type_of_manure</b>                              | 85                  | Cattle               | 59               | No                                                | 0.035                  |
|                                                    |                     |                      |                  | Sheep                                             | 0.024                  |
|                                                    |                     |                      |                  | Two third cattle One third sheep                  | 0.212                  |
|                                                    |                     |                      |                  | Ammonitrate N:33.5%                               | 0.294                  |
|                                                    |                     |                      |                  | Mineral fertilizer 12-12-17                       | 0.212                  |
| <b>Kind_of_fertilizers</b>                         | 85                  | Ammonitrate N:33.5%  | 25               | Mineral fertilizer 12-12-17 + Ammonitrate N:33.5% | 0.024                  |
|                                                    |                     |                      |                  | Mineral fertilizer 12-12-17 + Urea N:46%          | 0.024                  |
|                                                    |                     |                      |                  | Non                                               | 0.294                  |
|                                                    |                     |                      |                  | Urea N:46%                                        | 0.153                  |
|                                                    |                     |                      |                  | Cereals                                           | 0.435                  |
|                                                    |                     |                      |                  | Cereals + Herbs                                   | 0.012                  |
|                                                    |                     |                      |                  | Cereals + Vegetables                              | 0.165                  |
| <b>Crop_rotation</b>                               | 85                  | Cereals              | 37               | Cereals + Vegetables + Legumes                    | 0.024                  |
|                                                    |                     |                      |                  | Cereals + Vegetables + Legumes + Fallow           | 0.012                  |
|                                                    |                     |                      |                  | Cereals + leguminous                              | 0.318                  |
|                                                    |                     |                      |                  | Fallow                                            | 0.035                  |
| <b>Chemical_treatment_against_alfalfa_diseases</b> | 85                  | Yes                  | 38               | No                                                | 0.156                  |
|                                                    |                     |                      |                  | Yes                                               | 0.844                  |

Traditional and modern agriculture in the Moroccan agronomic system are undergoing degradation processes due to improper use of tillage practices tools. In contrast the transient beneficial effect of tillage, the sustainability of the adverse effects of conventional tillage systems destroys the environment [7]. The application of cattle manure to acid soils increases the pH, as it is attributed to buffering of organic acids and bicarbonates in cattle manure. The available mineral P, N, K, Mg and Ca increased immediately after manure application, and the available P and K remained significantly higher in modified soil by manure compared to unmodified soil [8]. The cattle manure increases in pH, contents of organic C, and total N and P. Also an increased microbial biomass and dehydrogenase and phosphatase activities of soil which gives an indication of the increase in soil fauna activity [9]. Alfalfa which is cultivated in soils polluted by heavy metals could tolerate high concentrations of Cd, Zn and Pb in the soil and had very good growth performance. Also, sheep manure also increases phytostabilization. Application of sheep manure can be helpful in increasing electrical conductivity and reducing the concentrations of DTPA-extractable metals in the soil [10], and can also reduce the effects of saline irrigation and increase the productivity of alfalfa [11]. The alfalfa and clover fix nitrogen and can reduce dependence on organic fertilizers and improve soil structure, for example the rotation between wheat and alfalfa [12]. Mineral nitrogen fertilization can be abandoned and arriving at a good yield of wheat by the cultivation of alfalfa for a period more than 3 years before the cultivation of wheat [13].

In order to meet the nutritional needs of humans to grow with quality and efficiency, it is known that the chemicals used in plant nutrition as well to increase soil efficiency and pest treatment are the most common practices, important for controlling the production and the quality of plant production, this non-reasonable use causes damage to the soil as the salinity, the accumulation of heavy metals, the eutrophication of water and the accumulation of nitrate and also pollution air by sulfur and nitrogen which they have dangerous effects on the environment like the greenhouse effect and consequently of the effects on the health and the existence of the human being [14]. The groundwater is exposed to contamination from pesticides and fertilizers used in agriculture [15]. Among strategies useful in reducing the use of fertilizers and pesticides are the productions of legumes, which fix atmospheric nitrogen by symbiosis with rhizobium nitrogen-fixing bacteria, in rotation with non-legume crops [16]. Also the combination of manure and straw increases both the yield and the microbiological activity of the soil also minimize of production input costs [17].

### 3.4. *Alfalfa cultivation Perspective*

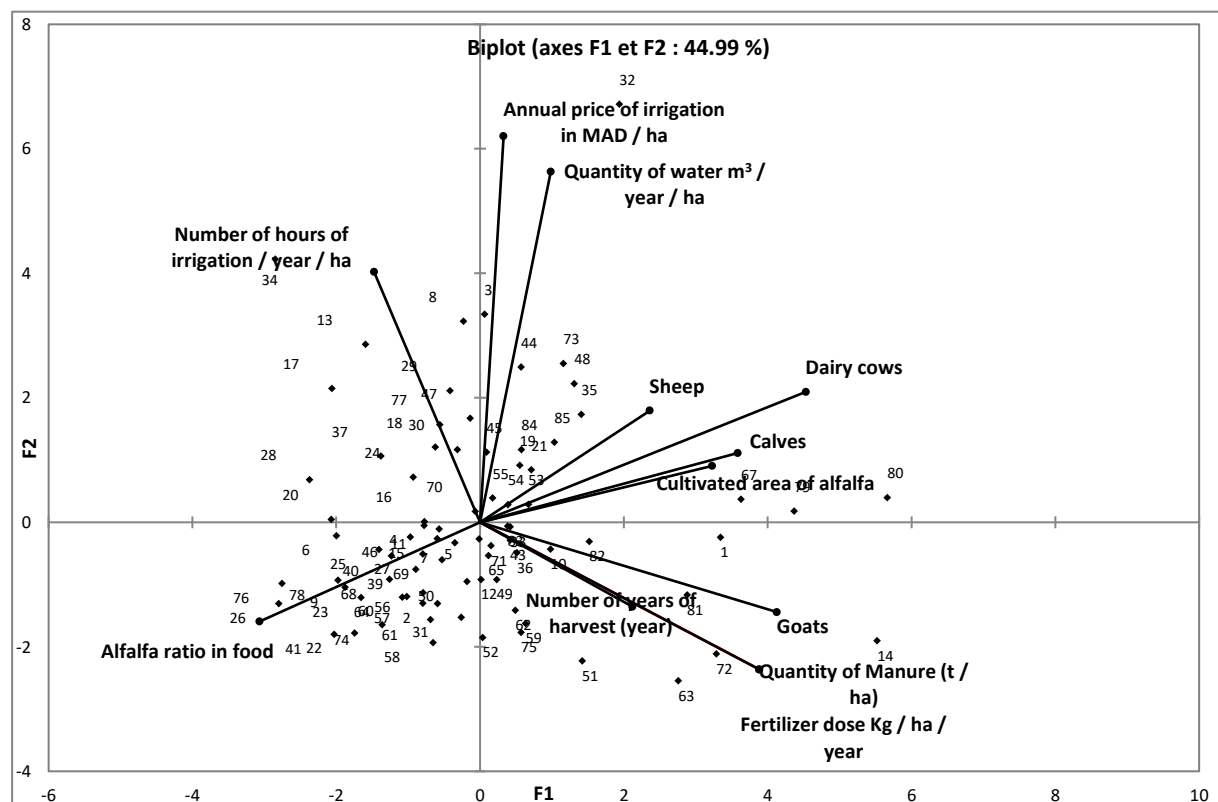
According to eighty-five questionnaires, only one person has another source of income, and four people will give up growing alfalfa, no one has answered yes regarding land abandonment or land conversion to real estate.

### 3.5. *Quantitative survey parameters*

In order to conclude the different correlations between the different quantitative variables we analyzed all the quantitative variables by PCA analysis (Figure 2). This analysis showed a strong positive correlation between the flock size and the area of cultivated alfalfa also between the quantity of fertilizers and with the numbers of the years of the successive harvest this graphical representation shows a negative correlation between the flock size and the incorporation of alfalfa in the feed ratio, that means when the size of the herd increases the quantity of alfalfa incorporated into the feeding system, other foods decreased in favor, also the graph shows a positive correlation between water consumption, cultivated area and the price. Although the price of m<sup>3</sup> changes according to the irrigation method. The PCA analysis shows that there is no relationship between water consumption and the use of organic or mineral fertilizers. Regarding the analysis of the range results of the individuals, it is observed that a group based mainly on a large ration of alfalfa compared to the other feed diet foods, this group is distinguished by a small herd size and an average consumption at low water since the cultivated area is low, the second group is characterized by a high consumption of

water and a low use of fertilizers, the third group is not based on alfalfa despite the cultivated area with alfalfa which is important but they do not support the large size of the herds.

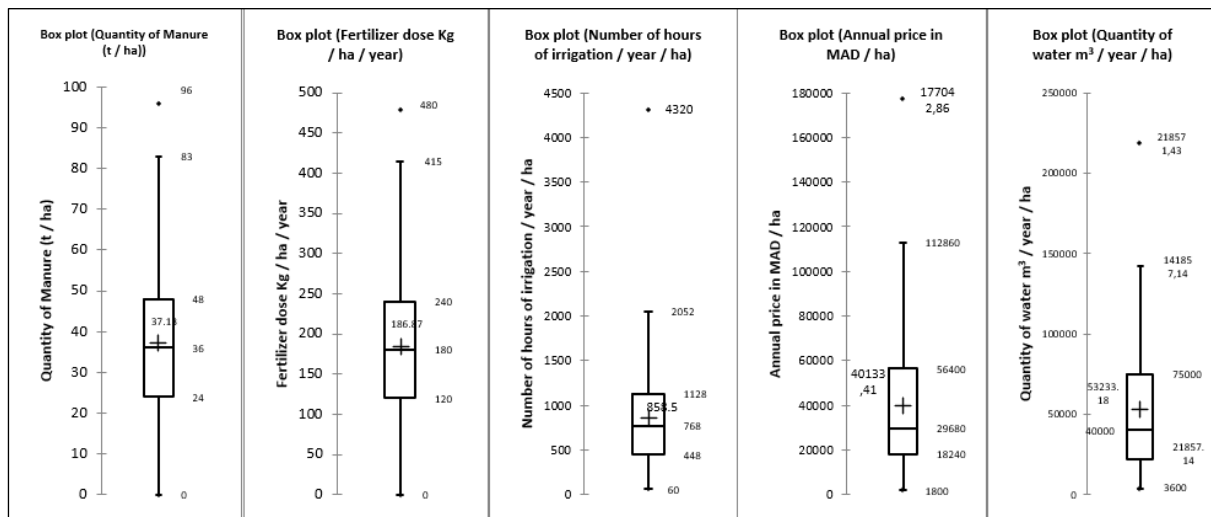
The area cultivated alfalfa in Morocco is appreciated by a quarter of the areas cultivated by fodder food in Morocco [18]. The incorporation of wheat straw in a diet which constitutes the corn silage acts better than a diet which is based on alfalfa straw since it has improved food intake and ruminal pH and also improves undigested neutral detergent fiber after 30 hours (uNDF30) and the ruminal function of high-yielding dairy cows, but may decrease feed efficiency [19]. The diet and its impact is not influenced by the change in composition but it is even influenced by the size of the same food for example dry matter intake (DMI), average daily gain (ADG) and the carcass weight[20].



**Figure 2.** Biplot graph of the principal component analysis (PCA) for the variation of following quantitative variables: Alfalfa ratio in food (%); Quantity of Manure (t/ha); Fertilizer dose (Kg/ha/year) ; Number of hours of irrigation (year/ha) ; Quantity of water (m<sup>3</sup>/year/ha) ; Annual price of irrigation in (MAD/ha) ; Number of years of harvest (year). ● Variable; ◆ Individual.

Box plots Figure 3. showed values of the application of extreme manure of the sample between 83t/ha and 96t/ha, the average use is about 37.13t/ha. Reconciling the median and the average shows statistical homogeneity and non-concentration of the data. The results of Fertilizer dose (Kg/ha/year) have shown strong annual uses of chemical fertilizers, 75% of the population uses an annual dose of chemical fertilizers greater than 120 Kg/ha/year. We also observe that 75% of the population with a number of hours less than 1128 hours/year/ha, a significant dispersion of the data for 25% of the individuals who stay. In accordance with the results of Box plots Quantity of water (m<sup>3</sup>/year/ha) and Annual price (MAD/ha), the statistical data are concentrated between 40000 m<sup>3</sup>/ha/year and 3600 m<sup>3</sup>/ha/year and 29680 MAD/ha/year and 1800 MAD/ha/year which represents 50% of the population, the other 50% of the population presents a significant dispersion of the data. The entire population

decreased irrigation in winter with proportions that have been taken into account in this present study. These observations can be explained by the heterogeneity in prices and debits according to the regions. Estimating means with a 95% confidence interval Table 3 showed the following results: The means of using chemical fertilizers is between 165.6 Kg/ha/year and 205.7 Kg/ha/year and Truncated mean at 5% 182.1 Kg/ha/year, and the means of the amount of Irrigation water is between 44066.8 m<sup>3</sup>/year/ha is 62399.9 m<sup>3</sup>/year/ha and Truncated mean at 5% is 49356.9 m<sup>3</sup>/year/ha.



**Figure 3.** BoxPlots which describe the parameters of irrigation and quantity of the organics and chemicals fertilizer

The water consumption by various plants is influenced by climate, water supply, soil and topography [21], [22]. This consumption is very important and very expensive hydraulically and economically for alfalfa given its continuous production and the long evapotranspiration times [23]. Insufficient irrigation negatively influences the quality and quantity of alfalfa since increases NDF and decreases the total productivity of alfalfa [24]. The increase in the number of years in the production of alfalfa is accompanied by a high consumption of water consumed from deep soil. Four years is the good duration of the cultivation of alfalfa which keeps both good production and low impact on the consumption of water from the deep soil [25]. The three-year alfalfa rotation followed by two Maize years increased the production of Maize and corn leaf area index, biomass, and yield by comparison with the Maize rotation alone and also decreased total evapotranspiration [26]. For continuous cultivation of alfalfa, it should be an additional of 4 to 5 tonnes / acre of farmyard manure to each cut. For the improvement of yield, it requires about 2 to 3 tonnes / acre of superphosphate per alternate year [27]. One-third-ton livestock manure and ammonitrate 33.5% fertilizer at 75 Kg/ha per application are most used on vegetable and leguminous crops in the Gharb region in Morocco [15]. The spread of pre-planned manure between 9 and 40 t / ha does not systematically result in more nitrogen in the soil or in the plant after 2 years of alfalfa [28].

**Table 3.** Estimate of the quantity of mineral fertilizers (A) and the quantity of water consumed per year (B) with a confidence interval of 95% with Average truncated at 5%.

| Feature A                    |                                       |             |                    |
|------------------------------|---------------------------------------|-------------|--------------------|
| Mean                         |                                       |             | Statistics         |
|                              |                                       |             | 185,67             |
|                              |                                       |             | Standard deviation |
|                              |                                       |             | 10,082             |
| Fertilizer dose (Kg/ha/year) | 95 % confidence interval for the Mean | Lower bound | 165,62             |
|                              |                                       | Upper bound | 205,72             |
|                              |                                       |             |                    |
|                              | 5 % truncated average                 |             | 182,05             |

| Feature B                                      |                       |             |            |                    |
|------------------------------------------------|-----------------------|-------------|------------|--------------------|
| Quantity of water<br>(m <sup>3</sup> /year/ha) | Mean                  |             | Statistics | Standard deviation |
|                                                |                       |             | 53233,17   | 4609,41            |
|                                                | 95 % confidence       | Lower bound | 44066,85   |                    |
|                                                | Interval for the mean | Upper bound | 62399,48   |                    |
|                                                | 5 % truncated average |             | 49356,89   |                    |

The application of manure up to 62.1t / ha for 4 years having positive effects on the biomass and on the dry weight also on Density of root mass and specific root length and on Plant available nitrogen compared to mineral fertilization with all applied doses arriving at N 210 kg / ha, P 300 kg / ha and K 150 hg / ha [29]. The application of high concentration of manure (336 kg N ha<sup>-1</sup>) had a negative effects on the productivity of alfalfa, Soil Nitrate and on nitrate in soil water over a long period [30]. The toxic effects of high doses of fertilizers reacted differently depending on the type of soil [31].

#### 4. Conclusion and perspectives

This survey showed the importance of the cultivation of alfalfa for the agricultures of the region and the problems linked to this crop and the bad cultural practices applied by agriculture such as the heavy dependence on plowing and the unreasonable uses of water resources. and mineral fertilizers, which firstly requires a study of the soil and the remaining water resources in order to describe the situation and the current state of these resources and to make decisions in order to save the situation, secondly, to seek other crops which do not consume water and are not sensitive to pests with the same positive impact or find control methods and water management methods in order to continue the cultivation of this fodder food.

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#### 6. References

- [1] R. Harbouze, J.-P. Pellissier, J.-P. Rolland, and W. Khechimi, "Rapport de synthèse sur l' agriculture au Maroc," p. 105, 2019, [Online]. Available: <https://hal.archives-ouvertes.fr/hal-02137637/document>.
- [2] E. Short, "The importance of forage," *Equine Heal.*, vol. 2018, no. 40, pp. 14–15, 2018, doi: 10.12968/eqhe.2018.40.14.
- [3] B. Boller, U. K. Posselt, and F. Veronesi, *Fodder Crops and*. 2010.
- [4] D. Verner *et al.*, "Climate Variability, Drought, and Drought Management in Morocco's Agricultural Sector," 2018.
- [5] G. Lardy and V. Anderson, "Alternative Feeds for Ruminants," *Ndsu*, vol. 1182, no. May, pp. 1–24, 2003.
- [6] K. P. Kulkarni, R. Tayade, S. Asekova, J. T. Song, J. G. Shannon, and J.-D. Lee, "Harnessing the potential of forage legumes, alfalfa, soybean, and cowpea for sustainable agriculture and global food security," *Front. Plant Sci.*, vol. 9, p. 1314, 2018.
- [7] R. Mrabet, R. Moussadek, A. Fadlaoui, and E. van Ranst, "Conservation agriculture in dry areas of Morocco," *F. Crop. Res.*, vol. 132, pp. 84–94, 2012, doi: 10.1016/j.fcr.2011.11.017.
- [8] J. K. Whalen, C. Chang, G. W. Clayton, and J. P. Carefoot, "Cattle Manure Amendments Can Increase the pH of Acid Soils," *Soil Sci. Soc. Am. J.*, vol. 64, no. 3, pp. 962–966, 2000, doi: 10.2136/sssaj2000.643962x.
- [9] J. A. Parham, S. P. Deng, W. R. Raun, and G. V. Johnson, "Long-term cattle manure application in soil. I. Effect on soil phosphorus levels, microbial biomass C, and dehydrogenase

- and phosphatase activities,” *Biol. Fertil. Soils*, vol. 35, no. 5, pp. 328–337, 2002, doi: 10.1007/s00374-002-0476-2.
- [10] Z. Elouear, F. Bouhamed, N. Boujelben, and J. Bouzid, “Application of sheep manure and potassium fertilizer to contaminated soil and its effect on zinc, cadmium and lead accumulation by alfalfa plants,” *Sustain. Environ. Res.*, vol. 26, no. 3, pp. 131–135, 2016, doi: 10.1016/j.serj.2016.04.004.
- [11] S. Mbarki *et al.*, “Performance of Medicago sativa Grown in Clay Soil Favored by Compost or Farmyard Manure to Mitigate Salt Stress,” *Agronomy*, vol. 10, no. 1, pp. 1–14, 2020, doi: 10.3390/agronomy10010094.
- [12] W. J. Nauta, G. J. Burgt, and T. Baars, “Partner farms: a participatory approach to collaboration between specialised organic farms,” in *Designing and testing crop rotations for organic farming. Proceedings from an international workshop*, 1999, pp. 149–158.
- [13] A. R. Safahani Langeroodi, E. Campiglia, R. Mancinelli, and E. Radicetti, “Can biochar improve pumpkin productivity and its physiological characteristics under reduced irrigation regimes?,” *Sci. Hortic. (Amsterdam)*, vol. 247, no. September 2018, pp. 195–204, 2019, doi: 10.1016/j.scienta.2018.11.059.
- [14] S. Savci, “An agricultural pollutant: chemical fertilizer,” *Int. J. Environ. Sci. Dev.*, vol. 3, no. 1, p. 73, 2012.
- [15] B. Marouane *et al.*, “Impact of agricultural practices on groundwater quality: Case of Gharb region-Morocco,” *J. Mater. Environ. Sci.*, vol. 5, no. Suppl. 1, pp. 2151–2155, 2014.
- [16] J. E. Fox, J. Gullledge, E. Engelhaupt, M. E. Burow, and J. A. McLachlan, “Pesticides reduce symbiotic efficiency of nitrogen-fixing rhizobia and host plants,” *Proc. Natl. Acad. Sci.*, vol. 104, no. 24, pp. 10282–10287, 2007.
- [17] E. Liu *et al.*, “Long-term effect of chemical fertilizer, straw, and manure on soil chemical and biological properties in northwest China,” *Geoderma*, vol. 158, no. 3–4, pp. 173–180, 2010, doi: 10.1016/j.geoderma.2010.04.029.
- [18] A. Bouizgaren *et al.*, “Assessment of summer drought tolerance variability in Mediterranean alfalfa (*Medicago sativa* L.) cultivars under Moroccan fields conditions,” *Arch. Agron. Soil Sci.*, vol. 59, no. 1, pp. 147–160, 2013, doi: 10.1080/03650340.2011.606216.
- [19] A. Kahyani *et al.*, “Performance of dairy cows fed diets with similar proportions of undigested neutral detergent fiber with wheat straw substituted for alfalfa hay, corn silage, or both,” *J. Dairy Sci.*, vol. 102, no. 12, pp. 10903–10915, 2019, doi: 10.3168/jds.2019-16869.
- [20] A. E. Francisco *et al.*, “Effects of alfalfa particle size and starch content in diets on feeding behaviour, intake, rumen parameters, animal performance and meat quality of growing lambs,” *Meat Sci.*, vol. 161, p. 107964, 2020, doi: 10.1016/j.meatsci.2019.107964.
- [21] H. F. Blaney and W. D. Criddle, *Determining consumptive use and irrigation water requirements*, no. 1275. US Department of Agriculture, 1962.
- [22] H. El Moussaoui and L. Bouqbis, “Interactive Effect of Biochar and Bio-Compost on Starting Growth and Physiologic Parameters of Argan,” *Sustainability*, vol. 14, no. 12, p. 7270, 2022.
- [23] C. C. Sheaffer, C. B. Tanner, and M. B. Kirkham, “Alfalfa Water Relations and Irrigation,” no. 29, pp. 373–409, 2015, doi: 10.2134/agronmonogr29.c11.
- [24] S. Jafarian, M. R. Chaichi, and M. Moradi-Ghahderijani, “Effects of surfactant and limited irrigation on forage yield and quality of alfalfa (*Medicago sativa* L.),” *Aust. J. Crop Sci.*, vol. 10, no. 1, pp. 76–80, 2016.
- [25] H. Yang, S. Li, H. Sun, W. Wang, and F. Zhao, “Effects of substrate material on plant growth and nutrient loss,” *Polish J. Environ. Stud.*, vol. 27, no. 6, pp. 2821–2832, 2018, doi: 10.15244/pjoes/81069.
- [26] T. Sun and Z. Li, “Alfalfa-corn rotation and row placement affects yield, water use, and economic returns in Northeast China,” *F. Crop. Res.*, vol. 241, no. June, p. 107558, 2019, doi: 10.1016/j.fcr.2019.107558.
- [27] R. Rashmi and V. Manisha Sarkar, “Cultivation of alfalfa (*Medicago sativa* L.),” *Anc. Sci. Life*,

- vol. 17, no. 2, p. 117, 1997.
- [28] M. A. Schmitt, C. C. Shaeffer, and G. W. Randall, "Preplant manure and commercial P and K fertilizer effects on alfalfa production," *J. Prod. Agric.*, vol. 6, no. 3, pp. 385–390, 1993.
- [29] V. Vasileva and O. Kostov, "Effect of mineral and organic fertilization on alfalfa forage and soil fertility," *Emirates J. Food Agric.*, vol. 27, no. 9, pp. 678–686, 2015, doi: 10.9755/ejfa.2015.05.288.
- [30] J. Daliparthi, S. J. Herbert, and P. L. Veneman, "Dairy manure applications to alfalfa: Crop response, soil nitrate, and nitrate in soil water," *Agron. J.*, vol. 86, no. 6, pp. 927–933, 1994, doi: 10.2134/agronj1994.00021962008600060001x.
- [31] J. E. Ells and A. E. McSay, "Allelopathic Effects of Alfalfa Plant Residues on Emergence and Growth of Cucumber Seedlings," *HortScience*, vol. 26, no. 4, pp. 368–370, 2019, doi: 10.21273/hortsci.26.4.368.